

AI: from experimentation to implementation?

Generative AI—use cases and election risks



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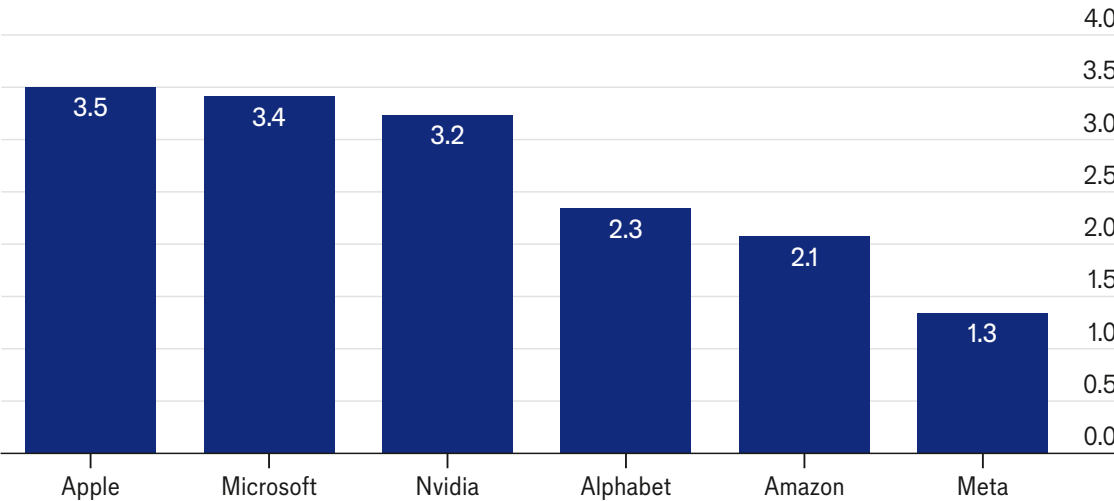
AI: from experimentation to implementation?

- The launch of ChatGPT and similar services has democratised the use of artificial intelligence (AI) in many industries, and companies are now experimenting with and implementing the technology.
- The main use cases so far for generative AI have been to improve operational efficiency, enhance innovation and support customer service through using chatbots.
- Most businesses still rely on non-generative AI, however, and it remains important for them to understand the limitations of generative AI technology, such as hallucinations.
- The biggest impact in the short term will be felt during democratic elections, particularly in countries with a polarised electorate, a fragmented information ecosystem and global influence (including the US).
- As AI implementation continues, sustainability will become a major barrier, given that generative AI systems use a lot of electricity and have a large carbon footprint.

Since OpenAI (US) launched ChatGPT in November 2022, AI, especially generative AI, has been driving investment for corporate entities. Nvidia, Microsoft and Apple (all US) have all been racing to be the most valuable companies in the world, with Nvidia leveraging its chips, Microsoft its cloud, and Apple its iPhone and other hardware. They are all looking to create a compelling ecosystem, even though this may put them even further in the crosshairs of regulators. But is progress in AI moving the technology from an experimentation to an implementation phase? This white paper will look at how companies have been using generative AI to date, and whether AI-generated content has had a major impact on elections so far in 2024.

Who will come out on top?

Market cap as at July 9th 2024; US\$ bn



Source: Companies; EIU.

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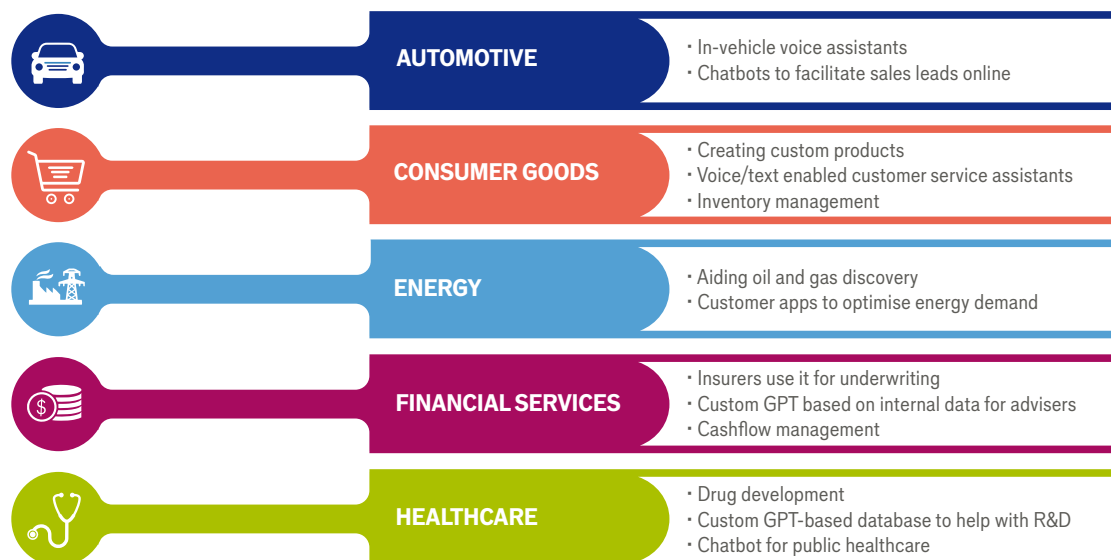
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How are companies using generative AI?

Companies in multiple industries have been experimenting with generative AI. Although the technology is new, two of the three major use cases across sectors are typical of many digital transformation strategies. The first is to **improve operational efficiency**, either in terms of increasing productivity or improving manufacturing and operational processes, often to cut costs. In the tech sector, this often includes using AI to speed up coding, or it can optimise internal processes such as training. Companies are also looking at technology to drive new revenue streams: generative AI is **driving innovation** across multiple sectors such as energy, financial services or healthcare, for example by making it easier to digest and analyse research papers. It is also used in **customer services** or to create better client-facing customer experiences. As the technology runs on large language models and is capable of dealing with vast amounts of text (as well as other media such as images), this is an area where development will continue across several industries, not just to interact with customers, but also for marketing purposes.

Companies use generative AI to drive innovation, efficiency and improve customer service



Source: Companies; EIU.

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In the **automotive** sector, Volkswagen (Germany) and Kia (South Korea) are among the vehicle-makers using generative AI to launch chatbots that allow the driver to control vocally various features within the vehicle (such as infotainment, navigation and air conditioning) to improve the driving experience. Chatbots are also employed for marketing and advertising purposes. Mercedes Benz (Germany) has built a chatbot tool (through a US social media site, Facebook) to help prospective buyers to customise their preferred vehicle and book a test drive, and Renault (France) has created a conversational advertising campaign to challenge car buyers' preconceptions about EVs.

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In the **consumer** sector, generative AI is offering some efficiency gains. Sainsbury's, a UK supermarket chain, has partnered with Microsoft to streamline workflows, and a US retailer, Walmart, has created an AI tool that allows in-store workers to get their work schedule or locate products via voice search. Generative AI has also been used for marketing campaigns. Carrefour (France) and Google (US) have partnered to create the Carrefour Marketing Studio, which allows the retailer to create campaigns in a few clicks while leveraging the huge amount of data—including the results of previous campaigns or product information—that it has at its disposal.

In the **energy** sector, generative AI is accelerating discovery and innovation. In May 2023 Shell (UK) announced a collaboration with SparkCognition, a US-based AI tools provider, to speed up the imaging and exploration of subsurface structures to help with oil discovery. In terms of efficiency, Cosmo Oil (Japan) has created a digital twin (digital representation of a physical asset) of its three refining facilities and is using generative AI to process historical data, helping the company to optimise processes such as site design, maintenance schedules and production estimates.

In **financial services**, generative AI has allowed the creation of new customer products; JP Morgan, a US bank, has used it to launch a cash flow management tool that scans financial documents and trains itself on the data patterns that it uncovers, identifying the most relevant data fields with every new document. This allows the tool to forecast future patterns of spending and deliver insights and recommendations. It offers greater efficiency, as seen in the launch of AI @ Morgan Stanley Assistant, which provides access to a vast collection of research resources and is designed to streamline daily tasks by automatically answering questions on various topics, including markets, internal procedures and investment recommendations.

In the **healthcare** sector, chatbots are being trialled to improve public healthcare services. Despite concerns about the ethical and privacy implications of using AI, the World Health Organisation (WHO) has taken the lead by launching a prototype chatbot, Sarah, which provides real-time information on topics like mental health, lifestyle and fitness choices. Merck (Germany) also launched its AIDDISON platform in December 2023 to expedite drug development by significantly reducing the time and resources required in pre-clinical trials; as it has been trained on more than 20 years of experimentally verified clinical datasets to identify the best possible compounds for a potential drug candidate.

Although investments into generative AI are increasing, 90% of AI usage within businesses still involves non-generative AI (also known as classical AI). Even where generative AI is used, many companies are still at the proof-of-concept stage: their experiments and projects will need to be scaled up over the medium to long term. But the launch of generative AI tools such as ChatGPT and Microsoft Copilot has democratised the use of the technology. This bodes well for future uptake, as long as companies and users understand its benefits as well as its limitations, such as its propensity to hallucinate (to generate misleading or incorrect information). Air Canada learnt this lesson when it launched a chatbot that gave wrong advice to consumers, for which the airline was liable. Examples like this show how costly it can be if AI is not implemented in the right way.

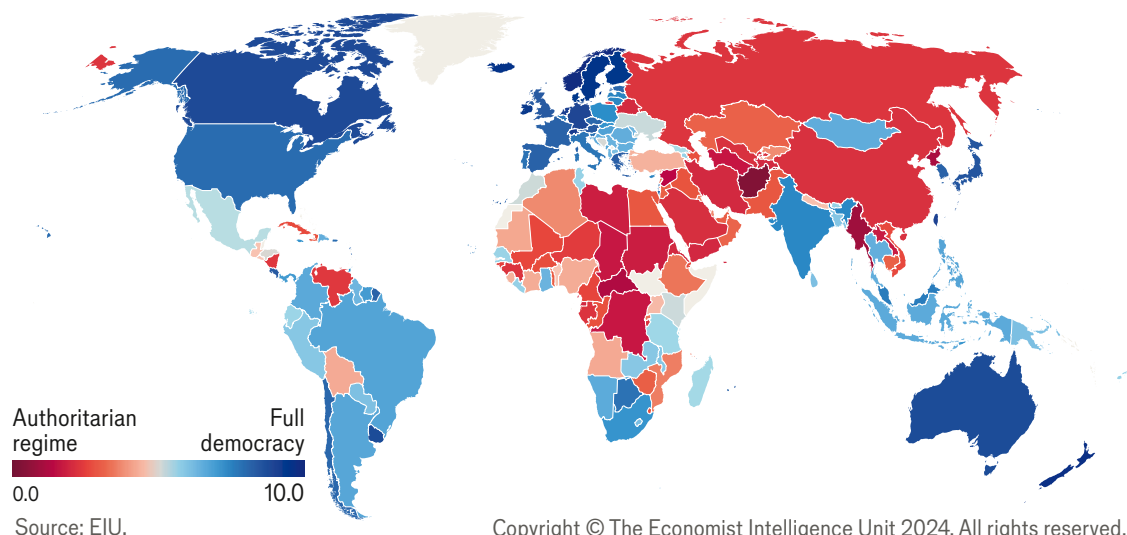
Can AI-generated content have an impact on elections?

Generative AI is not only being integrated into the corporate world; it is also infiltrating the political world. With 2024 being a year of multiple elections, and more than 4bn people called to vote, the impact of AI-generated content (both legitimate and fake) in election campaigns is only just emerging and will only increase.

One reason why AI usage for political purposes is accelerating is that the technology is becoming so much cheaper. Whereas content used to be expensive and costly to create, only a few thousand US dollars are now required to create multiple pieces of content. But the delivery will remain the most important element, particularly for fake content, because it affects how believable, clickable and shareable that content is.

For AI to have an impact on a country, we believe that four characteristics must be present: the country needs to be a democracy; it must have a polarised electorate as well as a fragmented information ecosystem; and it must have global influence.

Democracy Index 2023, global map by regime type



Democracies are most at risk. It may be obvious, but AI will only have an impact on elections in countries where polls are free and fair. As such, democracies are the regime most at risk, because of the openness of the political and information ecosystems. This will allow more misinformation to be published, which also allows foreign actors to take part in and influence the result of an election.

The need for polarisation. The more polarised a democracy is, the more likely it will be that AI-generated content will have an impact. It is easier for fake or doctored content to proliferate if there are two antagonistic camps that disagree on major issues and whose share of the vote could be broadly similar. Taking the UK as an example, the amount of AI-generated content seen during the 2024 general election was limited because of the lead that the Labour Party had in polls. However, greater interference is likely in the closely fought US election.

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A fragmented information ecosystem. The UK also benefits from a strong traditional media ecosystem, with a public broadcaster and well-established rivals leading the discourse. However, in many countries social media has become the primary source of information, especially among younger voters (who will soon become older voters). In France, National Rally (RN) leader, Jordan Bardella, has more than 1.5m followers on TikTok, increasing the party's popularity among younger voters. In May 2024 TikTok was temporarily banned in New Caledonia after rioting broke out, which the French government partly blamed on disinformation from China and Azerbaijan. Dis- and misinformation is more likely to spread via social than traditional media, and the increased popularity of the former means a greater likelihood of fake content.

A global influence. Although most AI-generated content is made domestically, in some countries it will have a strong foreign element. This is the case in the US, where Russia, China or Iran are very present. The country is highly polarised and has a fragmented information ecosystem, but most importantly it is the leader of the free world and the guarantor of the post-second world war liberal consensus. Influencing elections to weaken the US's role is critical to many of the country's adversaries, and this is why the US will be the bellwether as to the impact of AI-generated content in 2024.

Slovakia has demonstrated what can happen. In September 2023 Robert Fico won the country's parliamentary election, despite having been behind in the polls throughout. Over the last weekend before the elections an audio deepfake was released in which the leader of Progressive Slovakia, a rival party that had been ahead in the polls, was "heard" discussing election rigging. Slovakia fits all four of our criteria: it is a democracy (albeit a flawed one according to our Democracy Index); it is polarised between pro-Western and pro-Russia supporters; the deepfake was released at a time when the traditional media could not discuss the elections, meaning that it proliferated on social media; and Slovakia has global influence because of its membership of the EU and NATO.

It is difficult to make a direct link between a particular piece of AI-generated content and an electoral outcome. That outcome also does not need to be a victory for a specific party or candidate; the content can merely create a toxic environment. Even if AI-generated content does not have a major impact on elections this year, it may well do so in the future. Like cyber-security, this is a risk that cannot be prevented, but needs to be mitigated, as it will not disappear.

Sustainability will become a major barrier to AI adoption

AI will continue to be implemented in 2025, but we expect that the main focus will be on sustainability. The 2024 Electricity report from the International Energy Agency (IEA) noted that global electricity demand from data centres, driven by AI usage, could double between 2022 and 2026, adding to the grid the equivalent of Germany's national consumption. Regulators are already looking into the issue. In the US, the Artificial Intelligence Environmental Impacts Act of 2024 was proposed in March 2024 to assess and mitigate the environmental impact of AI; however, it is unlikely to become law before the November 2024 elections. The EU is further ahead, with the European Commission adopting the Energy Efficiency Directive (EED) in March 2024, which Germany has already implemented into law. From September 2024 data centre operators will be required to report on areas such as data volume, renewable usage or waste utilisation to reduce energy consumption and carbon emissions.

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The implementation and evolution of AI will be an ever ongoing process. The technology needs to be scaled up, and proponents need to move away from overoptimistic forecasts of artificial general intelligence (AGI) happening before the end of the decade. AI does not need to be perfect to have an impact. In fact, everyone should be aware that it is not perfect, and its usage will depend on the use case and require critical human oversight.

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